Properties of Matter

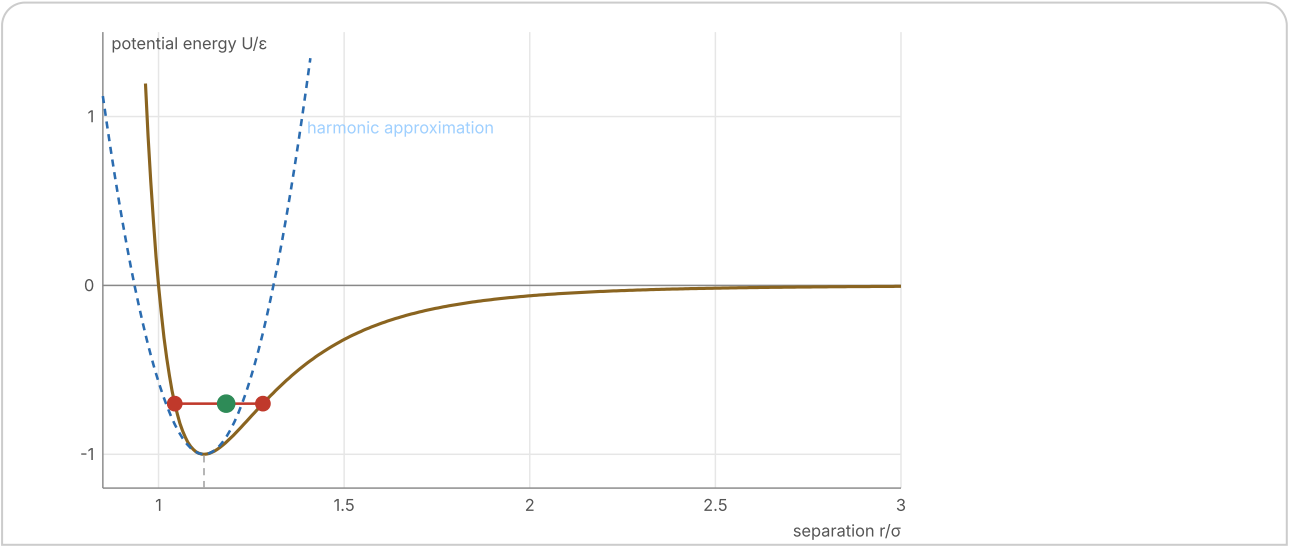
Why is steel stiff and rubber soft? Why does water rise in a thin glass tube while mercury sinks? Why is syrup viscous, and why does a gas not become more viscous when it is compressed? *Properties of Matter* was a classic course at British universities: it connects the measurable properties – elasticity, surface tension, viscosity, diffusion and heat conduction – with the forces between molecules. This page covers the subject with three interactive figures.

Forces between molecules

Almost all the mechanical and thermal properties of matter can be traced back to one curve: the potential energy between two molecules as a function of their separation. At long range the molecules attract each other weakly (van der Waals forces, due to fluctuating electric dipoles – Fritz London explained them quantum-mechanically in 1930). At short range they repel strongly, because the electron clouds cannot overlap (the Pauli principle). In 1924 John Lennard-Jones proposed a simple model:

$$U(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right], \qquad r_0 = 2^{1/6}\sigma$$

The depth of the curve ε determines how strongly the substance holds together (heat of vaporisation, boiling point, surface tension), its curvature at the bottom determines the stiffness (Young's modulus), and its asymmetry – it is steeper inwards than outwards – explains thermal expansion.



Kinetic theory of gases

In a gas the molecules are far apart and feel each other only in brief collisions. The pressure is due to the impacts on the walls. From Bernoulli, Clausius and Maxwell comes the result

$$pV = \tfrac{1}{3} N m \langle v^2 \rangle = N k_B T$$

The average distance a molecule travels between two collisions is the *mean free path*:

$$\lambda = \frac{1}{\sqrt{2}\,\pi d^2 n} = \frac{k_B T}{\sqrt{2}\,\pi d^2 p}$$

For air at room temperature and normal pressure $\lambda \approx 68$ nm – about 200 molecular diameters. A molecule collides about five billion times a second. In outer space or in a good vacuum pump the free path becomes kilometres long. See also Maxwell's speed distribution.

Transport in gases: viscosity, heat conduction and diffusion

When molecules fly from one layer of the gas to another, they carry momentum, energy and their own identity with them. This gives three transport phenomena with almost the same formula:

Phenomenon	What is transported	Kinetic theory
Viscosity	momentum	$\eta \approx \tfrac{1}{3} n m \langle v \rangle \lambda$
Heat conduction	energy	$\kappa \approx \tfrac{1}{3} n c_v \langle v \rangle \lambda$
Diffusion	particles	$D \approx \tfrac{1}{3} \langle v \rangle \lambda$

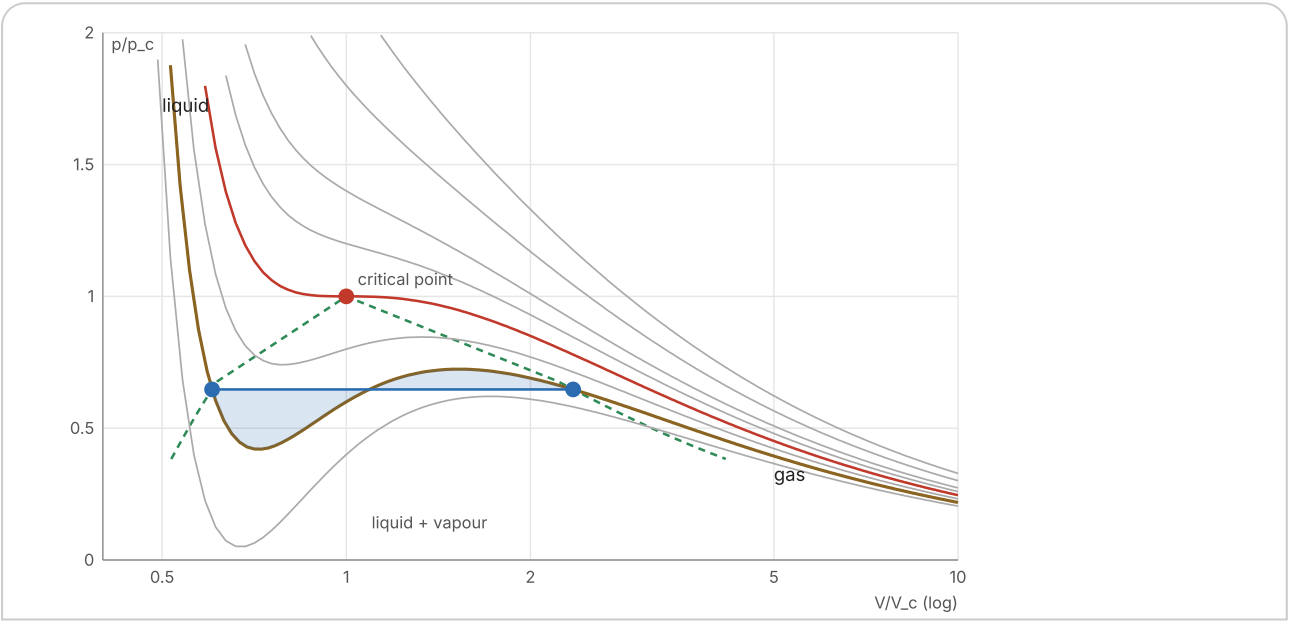
In 1860 Maxwell noticed a surprising consequence: since $\lambda \propto 1/n$, the density drops out of the formula for viscosity. The viscosity of a gas is independent of pressure! At first he did not believe it himself, but in 1866 he measured it with oscillating discs in an airtight box at his home in London – with his wife Katherine as assistant – and found it confirmed. He also found that the viscosity of gases increases with temperature, unlike that of liquids. The formula breaks down only when the pressure is so low that λ becomes comparable with the size of the container.

Real gases: van der Waals

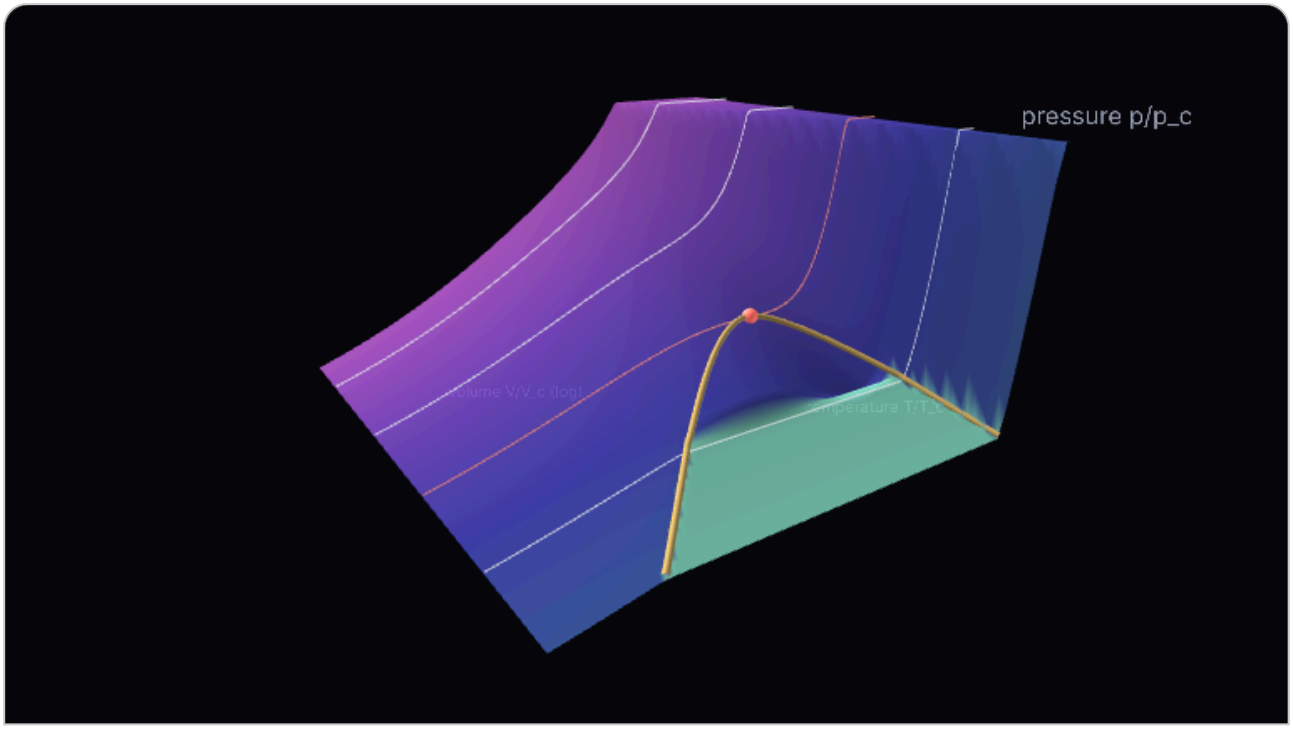
The ideal gas law assumes that the molecules are points with no forces between them. Johannes Diderik van der Waals corrected both in his doctoral thesis at Leiden in 1873:

$$\left(p + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

The constant b is the molecules' own volume, and a describes the attraction, which reduces the pressure on the walls. The equation predicts a *critical point*, above which there is no difference between liquid and gas, at $T_c = 8a/27Rb$, $p_c = a/27b^2$ and $V_c = 3nb$. Below T_c the isotherms develop a wiggle that is not physical; in 1875 James Clerk Maxwell showed that it must be replaced by a horizontal line placed so that the two areas are equal. Along that line liquid and vapour coexist. Van der Waals received the Nobel Prize in 1910. In 1908 his equation was Kamerlingh Onnes' guide to liquefying helium (see Superconductivity).



The van der Waals surface in 3D



Elasticity

In 1678 Robert Hooke found that the extension of a spring is proportional to the force. For a material this is expressed in terms of stress $\sigma = F/A$ and strain $\varepsilon = \Delta L/L$. In 1807 Thomas Young introduced what we call Young's modulus (the modulus of elasticity):

$$\sigma = E\,\varepsilon, \qquad G = \frac{\tau}{\gamma}, \qquad K = -V\frac{\Delta p}{\Delta V}, \qquad E = 2G(1+\nu) = 3K(1-2\nu)$$

An isotropic material has only two independent elastic constants; one typically uses Young's modulus E and Poisson's ratio ν, which tells how much thinner a rod becomes when stretched (typically 0.3 for metals, almost 0.5 for rubber, which does not change volume).

Material	E / GPa	Tensile strength / MPa	Poisson's ratio
Rubber	0.01–0.1	15	0.49
Wood (along the grain)	10	100	–
Bone	18	130	0.3
Aluminium	70	90–600	0.33
Glass	70	50	0.22
Steel	200	400–2000	0.29
Diamond	1050	–	0.1

From the Lennard-Jones curve one can understand the order of magnitude: Young's modulus is roughly the curvature at the bottom of the potential divided by the interatomic spacing. In theory materials should withstand stresses of about $E/10$, but real materials fail at much lower loads. In 1921 A.A. Griffith explained that small cracks concentrate the stress; in 1934 G.I. Taylor, Egon Orowan and Michael Polanyi showed that metals deform plastically because *dislocations* – line defects in the lattice – can glide. See also Continuum mechanics.

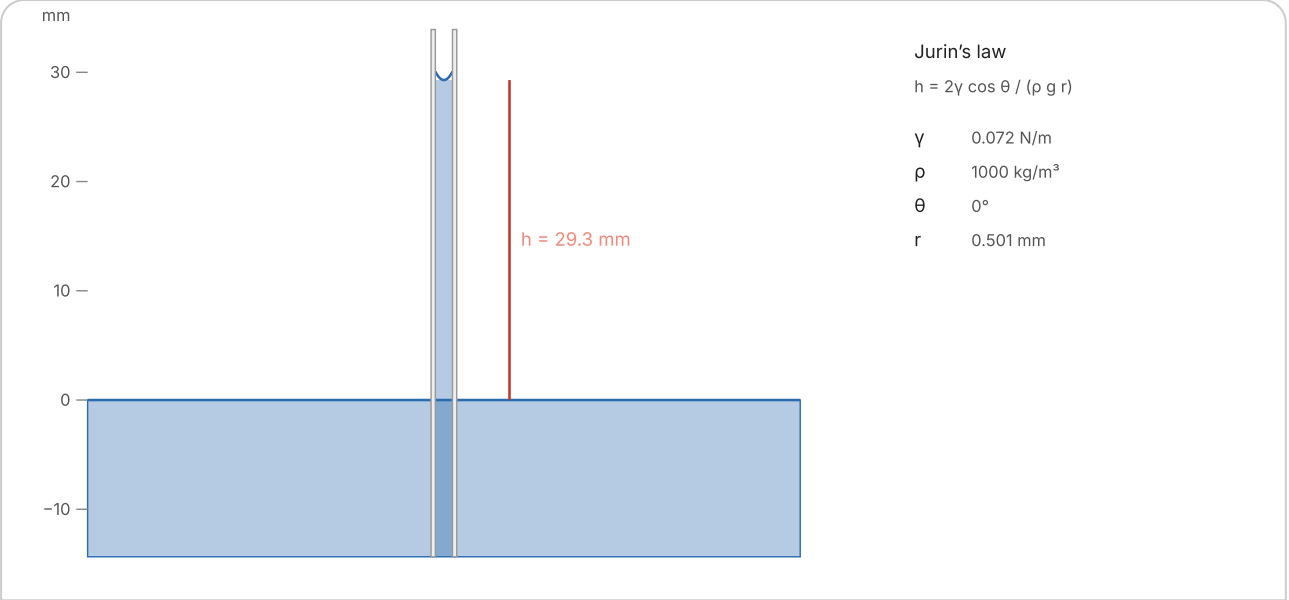
Surface tension and capillarity

A molecule inside a liquid is pulled equally in all directions, but a molecule at the surface is pulled inwards. Creating surface costs energy, and the liquid behaves as if it were covered by a stretched membrane. The surface tension γ is the energy per area or the force per length: 0.072 N/m for water at 25 °C, 0.022 N/m for ethanol and 0.485 N/m for mercury.

In 1805 Pierre-Simon Laplace and Thomas Young showed that the pressure inside a curved surface is higher than outside:

$$\Delta p = \gamma\left(\frac{1}{R_1}+\frac{1}{R_2}\right), \qquad h = \frac{2\gamma\cos\theta}{\rho g r}$$

The second formula, Jurin's law (James Jurin, 1718), gives the rise in a capillary tube of radius r. The angle θ between the liquid surface and the wall depends on whether the liquid "wets" the material: water on clean glass has $\theta \approx 0$, mercury has $\theta \approx 140^\circ$ – so mercury is depressed in a capillary tube, and its meniscus curves upwards. Capillarity lifts water in soil and in paper, and it is part of the explanation of how water rises in trees (together with the suction of evaporation).



Viscosity of liquids

In the *Principia* (1687) Isaac Newton assumed that the friction between two layers of a fluid is proportional to the difference in velocity between them: $\tau = \eta\, dv/dy$. Fluids that behave like this are called Newtonian. Two classic results follow:

$$Q = \frac{\pi r^4 \Delta p}{8 \eta L} \quad \text{(Hagen–Poiseuille)}, \qquad F = 6\pi\eta r v \quad \text{(Stokes)}$$

Jean Poiseuille, a physician, found in the 1840s from experiments with glass tubes that the flow rate grows with the fourth power of the radius – which is why even a slight narrowing of a blood vessel matters so much. In 1851 George Stokes derived the force on a sphere moving slowly through a viscous fluid. A sphere falling through a liquid reaches a constant *terminal velocity* $v = \frac{2}{9}(\rho_s - \rho_f) g r^2 / \eta$, and by measuring it one can determine the viscosity – the falling-ball viscometer. Robert Millikan used Stokes' law to find the size of the oil drops in his experiment that measured the charge of the electron (1909).

The viscosity of liquids falls sharply with temperature (syrup flows more easily when warm), because the molecules can more easily slip past each other. Non-Newtonian fluids such as cornflour in water become stiff when struck – see Cornflour and water. Flow on larger scales is treated in Fluid mechanics.

Substance (20 °C)	Viscosity / mPa·s
Air	0.018
Water	1.0
Olive oil	80
Glycerol	1400
Honey	2000–10,000
Pitch (the pitch drop experiment)	approx. $2 \cdot 10^{11}$

The pitch drop experiment in Brisbane, started in 1927, is the world's longest-running laboratory experiment: nine drops of pitch have fallen in almost a century.

Thermal expansion and heat conduction

Most substances expand when heated, because the atoms vibrate in an asymmetric potential (see the figure at the top). The linear expansion coefficient is about $12 \cdot 10^{-6}$ K⁻¹ for steel and $23 \cdot 10^{-6}$ K⁻¹ for aluminium. Water is an important exception: it is densest at 4 °C, so lakes freeze from the top.

Heat is conducted according to Fourier's law (1822) $\vec{q} = -\kappa \nabla T$ (see Fourier). In gases and insulators the heat is carried by molecules and lattice vibrations, in metals mainly by the free electrons – which is why good electrical conductors are also good conductors of heat (the Wiedemann–Franz law). Copper conducts heat at about 400 W/(m·K), glass 1, air 0.026 and aerogel 0.015.

Further reading

- D. Tabor: *Gases, Liquids and Solids, and Other States of Matter* (3rd ed., Cambridge 1991)
- F.H. Newman and V.H.L. Searle: *The General Properties of Matter* (5th ed. 1957)
- J.E. Gordon: *The New Science of Strong Materials* (Penguin 1968) and *Structures* (1978)
- Pierre-Gilles de Gennes, Françoise Brochard-Wyart and David Quéré: *Capillarity and Wetting Phenomena* (Springer 2004)

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